Juan Diego Jaramillo
judijasa@outlook.com

year 2014

Project 527

[Part 4]

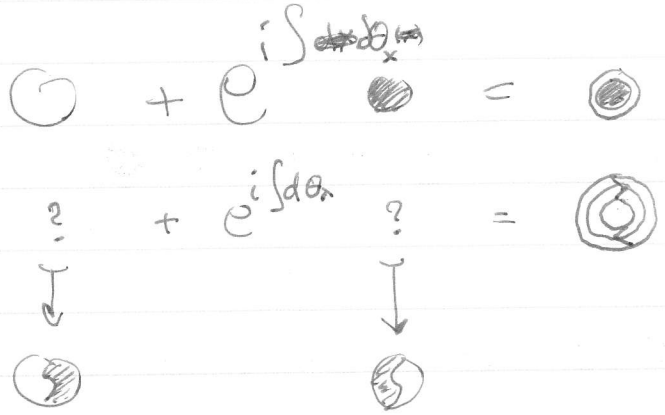
$[\cos\phi, \cos\alpha\theta] \gtrsim 0$, $[J_R^2, \underbrace{J_R J_L}] \gtrsim 0$

perturbation

I might have to deform space time to get

recall $\underbrace{g\,\partial_x\, g^{-1}}_{\partial_{\tilde{x}}}$

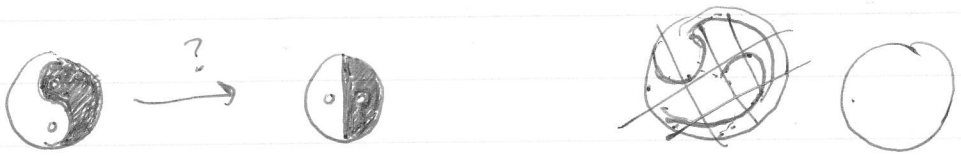
I could do ~~prop~~ postpone this treatment and first focus on showing that backscattering behaves as $\cos\alpha\phi$ (or $\cos\alpha\theta$) in the quantum band insulator... and then say this is not coincidence.



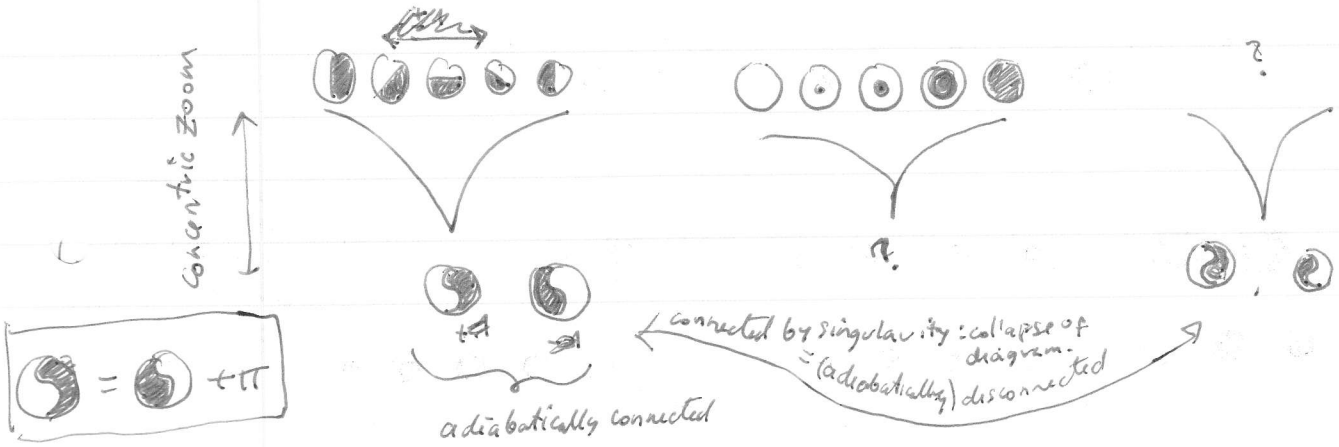
Meet the x-lattice:

... Develop the density wave description and then point out that $\cos\tilde{\alpha}\phi_\lambda$ (or $\cos\tilde{\alpha}\theta_\lambda$) is nothing but backscattering terms.

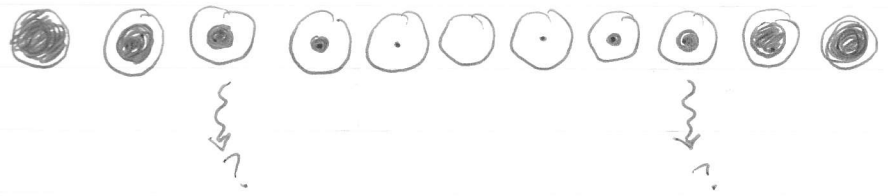
Better intro is to use ~~zoo~~ "concentric zoom" to show equivalence with traveling along x-space.



~~W~~ Perhaps when you write all supersymmetric hamiltonians you will understand how all this diagrams relate to each other.



Refinement of quantum solid



¿ why not ~~and~~ instead of and ?

Is not just a change of label, it creates a different edge state:

(Notice the relation between "tensor" product and diagramatics)

symmetric even for "−".

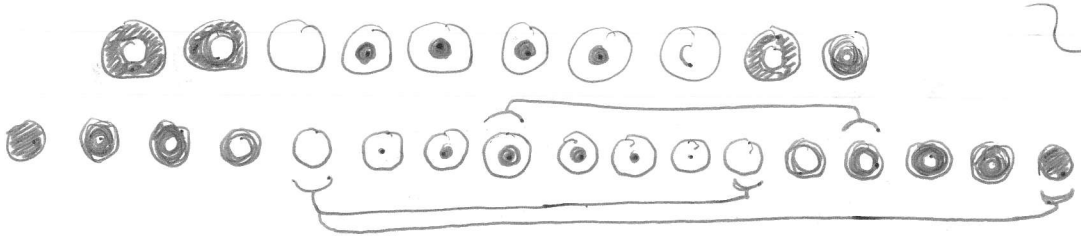
~~θ - x - lattice~~

~~θ - x - space~~

θ - x - lattice

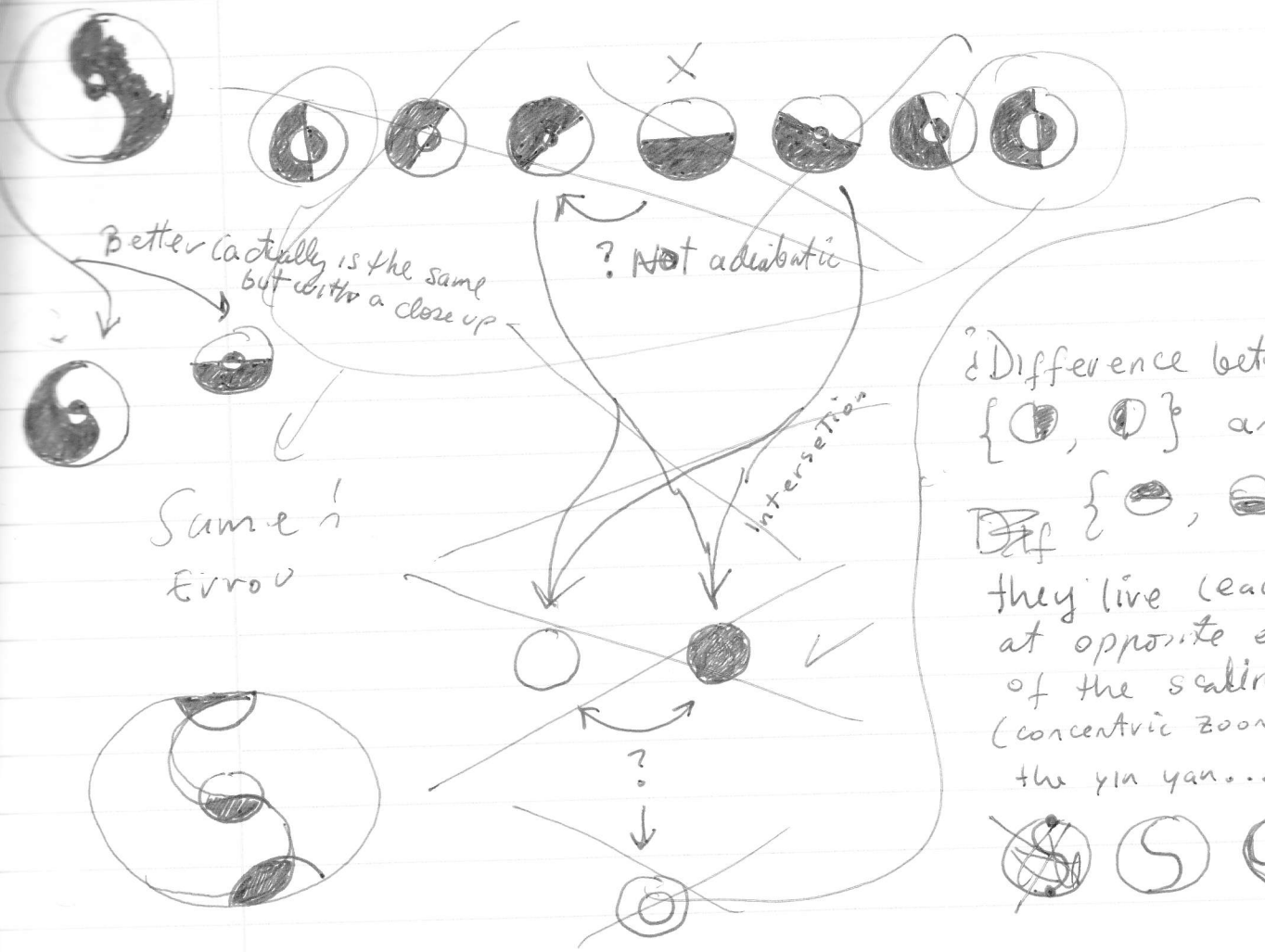
old version

"new" version



Creates color inversion of 'p' and 'h' after 2L, and for (p±h) after L → scaling imbalance or wrong diagram.

Connecting disconnected spots in yin yan? they always should be aligned



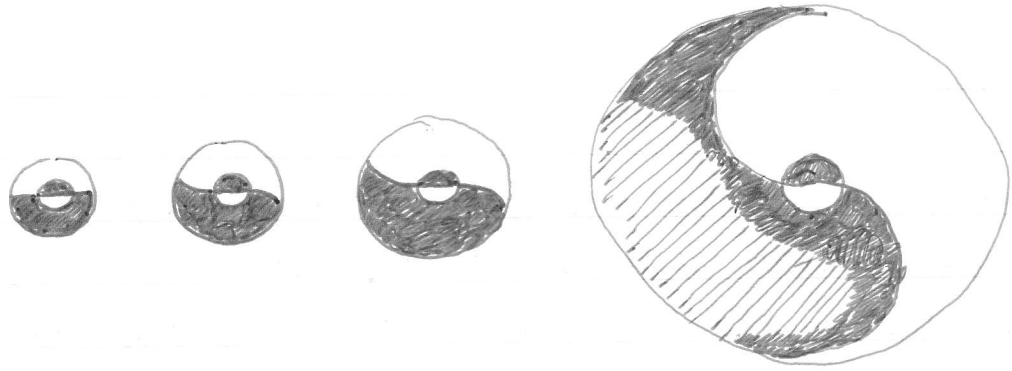
¿Difference between {, } and {, }, they live (each set) at opposite extremes of the scaling (concentric zoom) of the yin yan... or in a rotation of yinyan

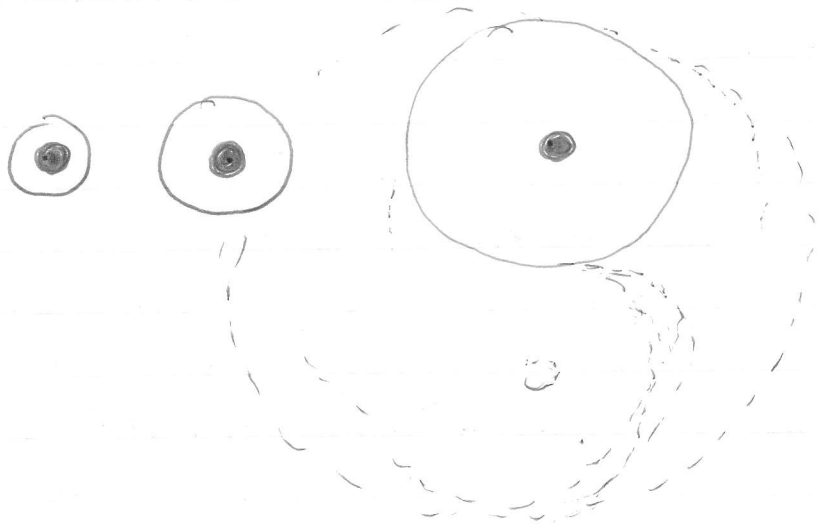
this diagram seems ~~not to favor~~ symmetric between ϕ and θ order → not good.

Concentric zoom ~ N (?)
↓
particle

Small concentric objects → "thorizont"

By generalizing (formalizing the def. in the most abstract way) the def. of edge states you might arrive at an homotopy theory. First try to construct the bulk as an 'integration' of edge states and so on.

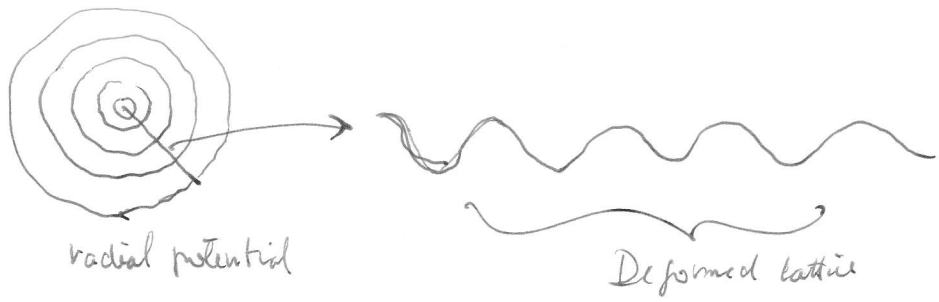




About experimental realization ...

My original scheme is in 1D. Sometimes is difficult to stabilize 1D deg. gas. Are there simple extensions to 2D or 3D? One possible model is ~~an extension~~ central potential (extension of mexican hat) ~~on~~

This is still '1D'



The signal of edge states might perhaps be extracted in this scheme ... the problem is that there is not an equal number of particles in each site (ring well) → how many? are you sure? nevertheless in the large radius limit $\frac{R-(R+r)}{R} = r/R \to 0$ (r is lattice const.)

Observation: In 1D edge states where 0D, in 2D edge states are 1D → remember articles from chinese scientist (Massardo times) where you had cutting with 'fines' strings objects ... strings are the extension of edge states to higher dimension is the connection between D and (D+1) ~~the bulk~~ ~~is~~ they are the key to define the homotopy.

How does ~~th~~ 'my' holographic' theory and other insights look in 2D [now you are ready to answer that as you have identified the edge states in 2D are 1D objects).

What if the edge states in 2D are like 1D i.e. s.t. they also have edge states (which are 0D)?

→ For the edge states of edge states is bulk, i.e. there is a map in which edge states look like bulk and bulk like edge states, ~~that~~ ('holographic' symmetry plays a key role), hence ~~I could introduce back~~ higher dimensions ~~as~~ could be hidden symmetries.

But would be suggesting that dim-(n-1) is dim-(n+1)... a sort of alternation ('néél like'). Can it be possible that lambda is also associated to ~~a~~ other spatial dimensions?

The periodicity of ~~λ-lattice~~ a config. in λ-lattice would be associated to dimensions of real space?

The lattice without deformation now plays the role of ~~the~~ a local system of coordinates. (frame)

To Do:

- $\begin{cases} \Delta_\alpha \theta & \text{(liquid)} \\ \Delta_{1/\alpha} \theta & \text{(solid)} \end{cases}$

a way to see in our scale: $x = n\alpha$

$$\Delta_{1/\alpha}\theta = \theta(n/\alpha + 1/\alpha) - \theta(n/\alpha) = \oint_x \delta\theta = \int_x^{x+L} \frac{\partial\theta}{\partial x}\, dx$$

where $\boxed{x_n = n\alpha}$

Now comeback to quantum solid and check if correction is

$\cos \alpha\theta(x)$ or $\cos\theta(n/\alpha)$ ~~$\cos\theta(x\alpha)$~~ $\to \cos\theta(x/\alpha^2)$

$x \to n\alpha$

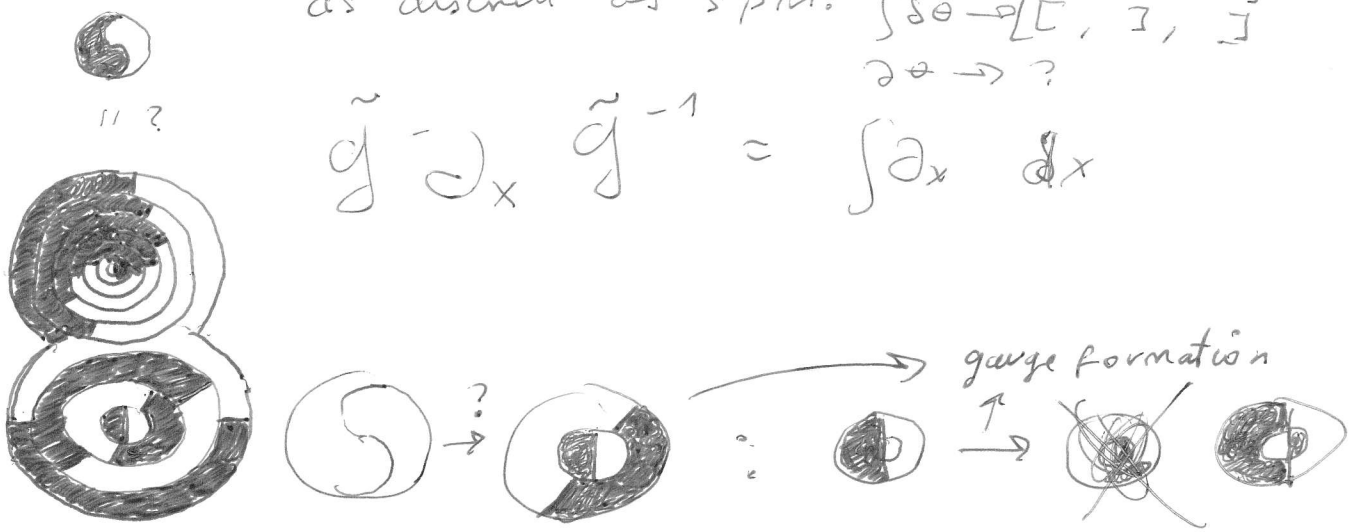
or <u>whatever</u>. Find out.

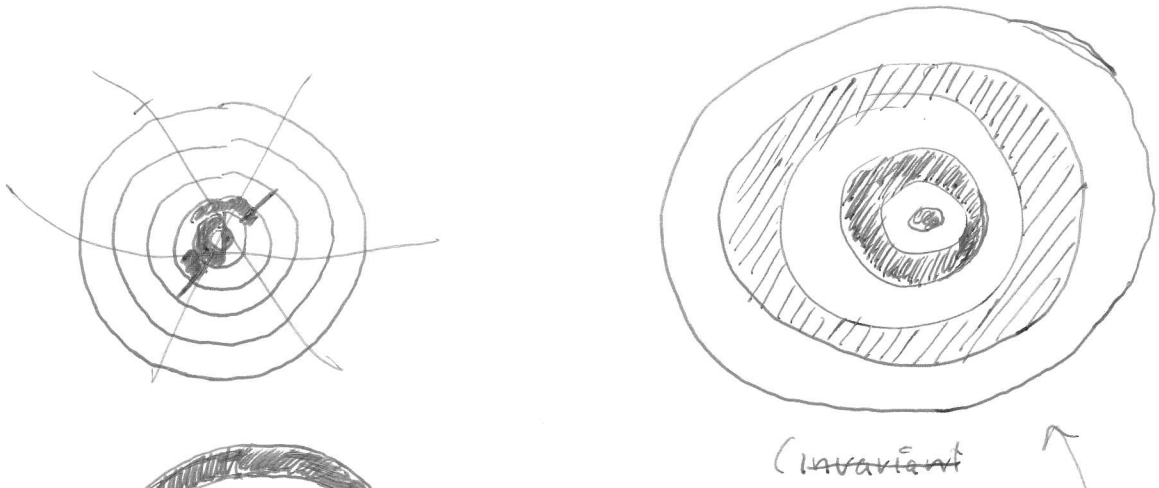
$$|\langle \Delta_{\alpha^{\pm 1}} \theta \rangle| = 1$$

* Need to understand this when space is as discrete as spin. $\int \delta\theta \to [\,[\,,\,],\,]$

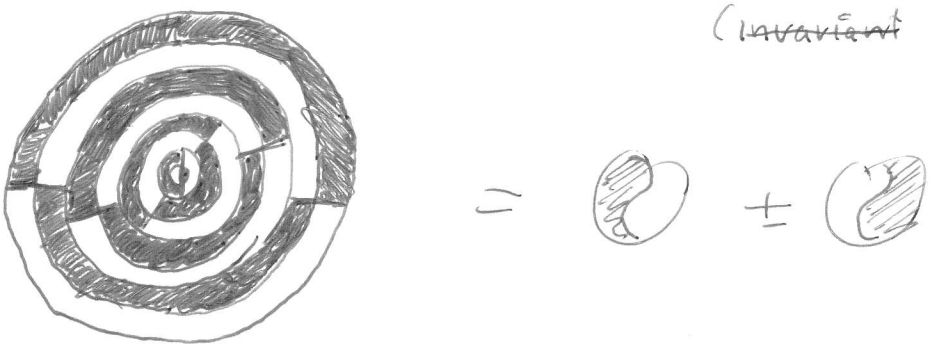
$\partial\theta \to ?$

$$\tilde{g}^{-1} \partial_x \tilde{g}^{-1} = \int \partial_x \, dx$$





(~~invariant~~



Gauge out the 1st liquid diagram to intro .

Careful: ~~inner concentric rings are~~ ~~the~~ using concentric rings for gauge and for scaling might not be consistent choose one (scaling)... no, it is consistent both are nothing but mapping of (next) lattice ~~next~~ sides into (next) concentric sites

inversion of concentric order is a rotation. ~~this cannot be~~ → if the object is alone. a rotation is an invariant operation. Unlike that creates a different object (Z/2) another 'need'

(2) → bandinsulator in x-lattice ... next. in x-space. how does the gauge in x-space (0+) looks like? like x-lattice. Hence in the same way solid has two concentric colors for its gauge, the liquid has two lattice colors for its gauge.

$$\phi(x+dx) = e^{i\delta\theta}\phi(x) = \phi + \delta\phi$$

$\Delta\theta = \pi$ solid

$\Delta\theta \to \delta\theta$ liquid

$\partial\theta = \pi$

$$\frac{\Delta\theta}{\Delta X} = \pi$$

$\Delta\theta$

$\Delta\theta(n\alpha) = \theta(n\alpha+\alpha) - \theta(n\alpha)$

solid:

$\sum_n \Delta\theta(n\alpha) = \theta(N\alpha) - \theta(0) = \pi$

$\theta_{n+1} - \theta_n$

$\theta(\alpha n+\alpha) - \theta(\alpha n)$

L/a $\qquad$ a/L

always $= \pi$ } what changes is α.

$\theta(\alpha n+\alpha) - \theta(\alpha n)$

$\int \delta\theta$ $\qquad$ $\partial\theta$

α_0 : <u>our</u> scaling of space-time.

$\partial\theta = \theta(\alpha n+\alpha) - \theta(\alpha n)$

$\int \delta\theta = \theta(L) - \theta(0) = \theta(\alpha^{-1}n + \alpha^{-1}) - \theta(\alpha^{-1}n)$

$\downarrow n=0$

$= \theta(\alpha^{-1}) - \theta(0)$

$\cancel{\Delta\theta(\alpha}$

$$[\Delta_\alpha \Theta]_\alpha = \frac{\partial \Theta}{\partial x}, \quad dx = \alpha$$

$$[\Delta_{1/\alpha} \Theta]_\alpha = \int \frac{\partial \Theta}{\partial x}\, dx$$ to be improved

$$[\Delta_\alpha \Theta]_{1/\alpha} = \int \frac{\partial \Theta}{\partial \lambda}\, d\lambda, \quad \cancel{d\lambda = 1/\alpha} \quad \lambda \overset{?}{=} \lambda(\alpha)$$

$$[\Delta_{1/\alpha} \Theta]_{1/\alpha} = \frac{\partial \Theta}{\partial \lambda},$$

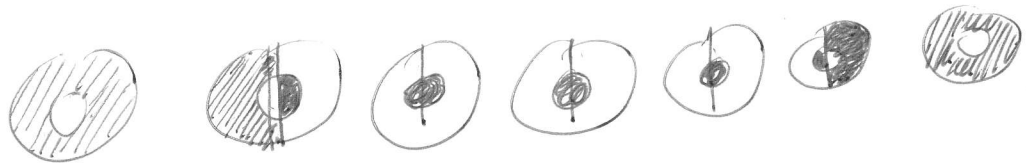
~~No, pu~~ No perhaps... $[\Delta_\alpha \Theta]_{dx} = \frac{\partial \Theta}{\partial x}$, $[\Delta_{1/\alpha}\Theta]_{dx} = \int \frac{d\Theta}{dx}\, dx$

$(\cancel{x, 1/x}) \leftrightarrow (\alpha, \frac{1}{\alpha})$ unitary

$[\Delta_{1/\alpha} \Theta]_{d\bar{x}} = \int \frac{\partial \Theta}{\partial \bar{x}}\, d\bar{x}$, $[\Delta_\alpha \Theta]_{d\bar{x}} = \frac{\partial \Theta}{\partial \bar{x}}$

$\bar{x} = n/\alpha$

Figure

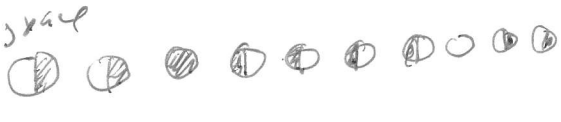


is redundant.

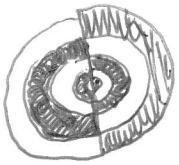
draw only: in x-space

in λ-space

and x-space



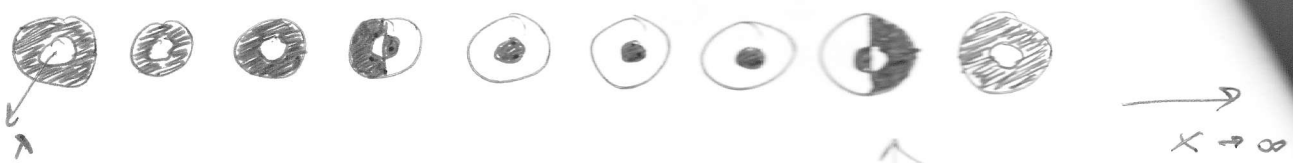
in λ-space



... Ah no, is not redundant because it recovers domain position symmetry!

[study ... θ-ord...

X-Space (actually is $X(\lambda)$-space)



λ-Space (actually $\lambda(X)$-space)

Note: but how to represent edge states in λ-space?

General observation: one cannot ~~deco~~ fully describe a quantum band insulator solely in X-space or λ-space, all you can do is ~~the~~ a sort of limit representation where the other degree of freedom is contracted to its minimal but not negligible form: parity: $\mathbb{Z}_2$

Notice that solid* = superconductor, liquid* = super-solid. the latter are exactly like solid and liquid but in $\psi_R \psi_L$ instead of $\psi_R^\dagger \psi_R$. Conductivity of liquid* is pairing density while single particle density ~~of the~~ is conductivity

$$\underbrace{\text{Conductivity}^*}_{\text{massless theory}} = \underbrace{\text{pairing density}}_{\text{density wave}}$$

$$\underbrace{\text{Single particle density}^*}_{\text{density wave}} = \underbrace{\text{Conductivity}}_{\text{massless theory}}$$

by understanding how $e \to 2e$ in superconductor using this language we might understand fractionalization in general.

- the gap of a liquid* (Supercond.) is the ~~confinement~~ dissociation energy of pairing
- the gap of a solid is the confinement energy of pairing.

[~~liquid*~~]* (liquid*)* = solid

$\langle \psi_R^+ \psi_R \rangle = 1 \quad , \quad \langle \psi_R \psi_L \rangle = 0$ liquid

$\langle \psi_R^+ \psi_R \rangle = 0 \quad , \quad \langle \psi_R \psi_L \rangle = 1$ solid*

((solid*)* = liquid)

$\langle \sum (-1)^n \psi_R^+ \psi_R \rangle = 1 \quad , \quad \langle (-1)^n \psi_R \psi_L \rangle$ liquid *

$$g_0 H_0 + \bar{g}_0 \bar{H}_0 + g_m H_m + \bar{g}_m \bar{H}_m$$

Liquid $\langle H_0 \rangle = 1$ else 0.

solid $\langle H_m \rangle = 1$ else 0.

liquid* $\langle \bar{H}_0 \rangle = 1$ else 0. $\rightarrow$ supersolid

solid* $\langle \bar{H}_m \rangle = 1$ else 0. $\rightarrow$ superconductor

Liquid* : $\langle (-1)^n \psi_R \psi_L \rangle = 1$ else 0.

solid* : $\langle \psi_R \psi_L \rangle = 1$ else 0.

$$(n\alpha,\ n/\alpha) = (x, \bar{x})$$

what is good of this is that again I get a space variable but in the opposite scaling sector. ~~what~~ But now I have to think further to arrive at a theory in α-space.

$$\cancel{(n e^{\alpha(m=1)}}$$

$$(\ldots, n_x e^{-2\lambda}, n_x e^{-\lambda}, n_x e^{\lambda}, n_x e^{2\lambda}, \ldots)$$

$$\cancel{(n_x e^{\alpha(m_\alpha=1)}}$$

$$\cancel{(n_x e^{\lambda(m_\lambda=1)}}, n_x e$$

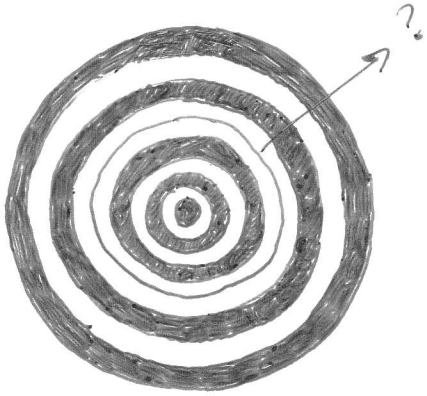
$$\boxed{\begin{aligned} dx &= \alpha \\ d\bar{x} &= 1/\alpha \quad \text{(very intriguing)} \end{aligned}}$$

$$\cancel{(n_x e^{\lambda}, n_x e^{-\lambda})}$$

$$\cancel{(\ldots, n_x e^{2\lambda}, n_x e^{\lambda}, n_x e^{-}}$$

$$\cancel{(n_x e^{-2\lambda}}$$

Edge-states of quantum Loci in λ-space



~~Alternative~~
Candidates:

1) Introduce geometric precision in the diagram and ~~and~~ add a lattice deformation in the radial axis. Only to disconnected rings are necessary (~~no~~ $\mathbb{Z}_2$ picture is preserved)

2) add many of this colony x-lattice and add the edge add corresponding generalization. this seems like a redundant scheme but it actually seems self consistent and ~~is not~~ it ~~es~~ doesn't excludes option (1) which could be understood as one of the minimal pictures.

Combined (1) and (2) Best option?

Estereographic #1

Notice that in deformed lattice, the transition (region of expected mixed states) is as ~~big~~ 'large' as pure color domains. One would say: 'how come if these are supposed to be edge states which we see as particles like instead of domain? Well, interestingly the size of edge states along the x-lattice is an observational effect since these objects are allways found at the horizont, they seem to be confined in a small region in the x-lattice similarly to how in the stereographic mapping of the plane, you see the horizont as a small region near ~~north~~ the north (or south in the other chart) pole.

Or maybe is the ~~bodomain~~ bulk which is an 'optical illusion' or ~~maby b maybe~~ might be both.

And maybe, when we 'see' a particle i.e. we observe directly an edge state (not in the horizont) is because we are in a different chart; like when we go from North to South pole charts; what appeared to be a horizont is now bulk and so on; but when we 'see' a particle with finite size, we are ~~not~~ perhaps ~~not fully~~ in an intermediate between south and north charts.

Important: set the ~~radius~~ radius of sphere to infinity s.t. when you look at ~~the~~ 'ecuatorial objects they will look of infinite size → bulk. Only then we recover a theory with infinite large and small objects.

I need to go in depth with the stereographic effects in ~~ed~~ observations. ~~Incorporate this mapping~~ ~~int~~ Generalize this mapping as to allow holographic symmetry: why some configurations look like ~~vacuum~~ ~~waves~~ bulk (renormalized vacuum) (~~renormalized vacuum~~) and others like ~~particles~~ (~~ed~~ ~~edge st~~ edge ~~stats~~ states (horizont).

(what is the opposite of horizont?)

~~Bulk~~

Case 1

Bulk
(~~waves~~)

Edge states
(~~particles~~)

Horizont

'center'

Case 2

Bulk

Edge states

Horizont

Center

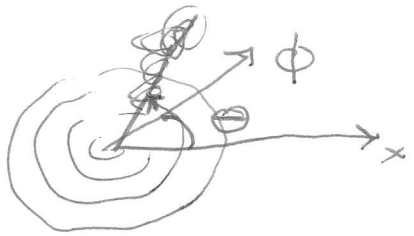
Case 1 → Case 2

→ One step along λ-lattice in Hamiltonian with 'holographic' symmetry.

$\phi \equiv \lambda$

$\partial_x \phi$ $\quad \partial_\phi X$ $\quad \partial_\phi X$

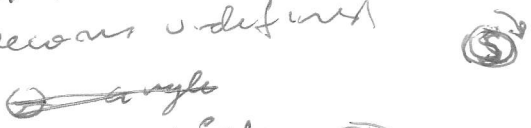
$\partial_x \theta$ $\quad \partial_\theta X$ $\quad \partial_\phi t$



fix x-position

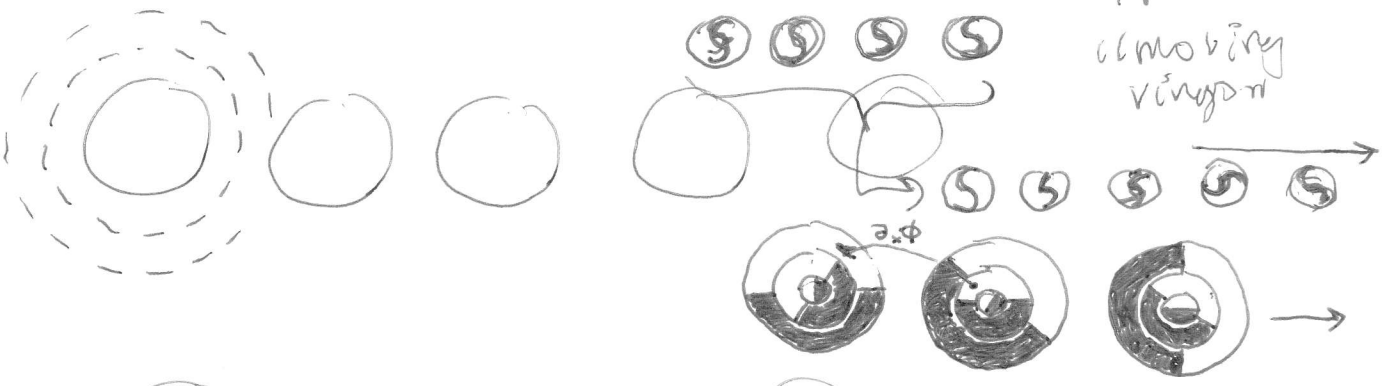
when a liquid becomes undefined

~~angle~~



inner ring expands.

outer ring rotates



the incompressibility of a liquid is due to this symmetry between x-space and λ-space configuration. (the symmetry has geometric precision)

Add area preserving ~~rate~~ radius rate to the λ-space representation and see if you get closer to yin yan; ~~in fact~~ revise the code, in case it is already there, take it away and see what happens.

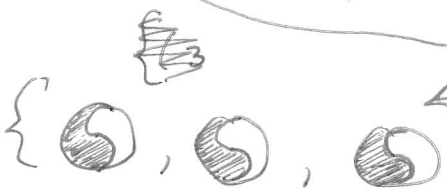
$\partial_x \phi$ is responsible for ~~transporting~~ expanding a ~~ring~~ given ring (with well defined θ) every x-step.

apply the same mapping to the solid you know what to expect.

mapping the liquid from x to T repr.

Letting radius grow as $r^{1/2}$ makes something slightly closer to yin yan but is not justified.

(x, λ) :



$$= \left\{ \begin{array}{l} \{(1,(3,4,5)), \\ (2,(2,3,4)), \\ (3,(1,2,3))\} \end{array} \right. = \left\{ \begin{array}{l} (1,3)(2,2)(3,1) \\ (3,2)(2,3)(1,4) \\ (3,3)(2,4)(1,5) \end{array} \right.$$

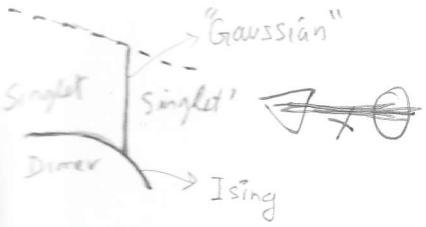
But λ-space is not simply the momentum space of a single particle because in the latter, the conformal dimension is preserved; in λ-space there is fractionalization of whatever ~~deg~~ discrete degree of freedom that is carried by 'momentum'. ~~is like~~ ~~dim X_n~~ is like in x-space when you allow ~~the existence of a multiplicity~~ several classic regularization parameters; ~~you cannot have the~~ a single particle has only one. The same with ~~lambda~~ λ-space: it allows having momentum states attached to different 'positions' (discrete) thanks to the many-body nature. ~~Usually~~ single particle quantum mechanics will not ~~fix position~~ allow this; it is like cheating on the uncertainty principle, but is possible because the uncertainty principle of many-body system is ~~def~~ not the 'simple sum' of single particle uncertainty principles.

singlets:

$\langle \cos\sqrt{4\pi}\,\theta_v \rangle \approx 1$

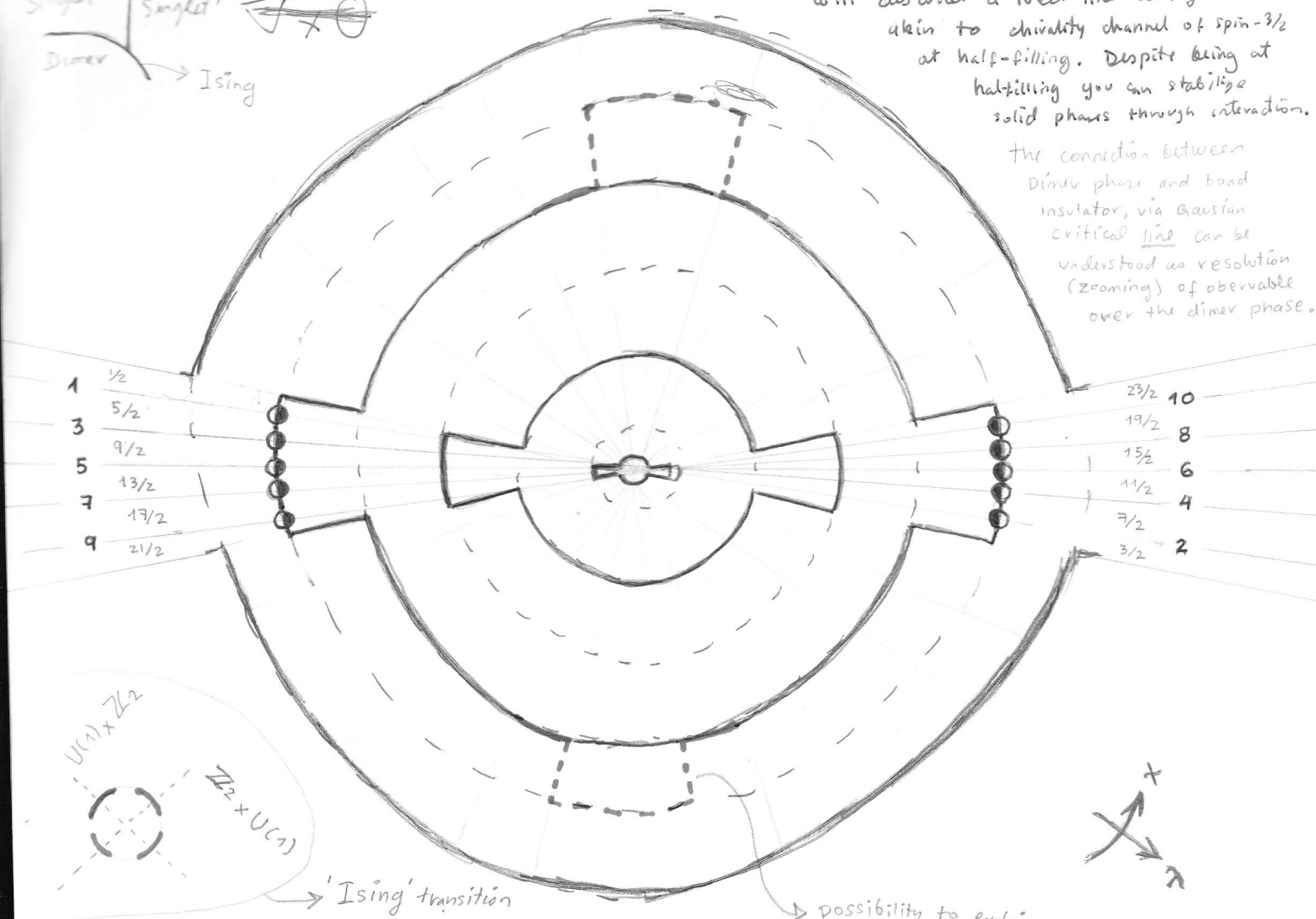
the angle in which the solid opens on the liquid to seem stable is smaller but for diagramatic purpose I ~~kept~~ ~~below~~ larger.

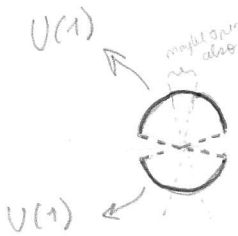
B.I.



I've changed conformal dimension, instead of 1/2 'steps' I use ~~only~~ unit 'steps'. this will describe a Neel like configuration akin to chirality channel of spin-3/2 at half-filling. Despite being at halfilling you can stabilize solid phases through interaction.

the connection between Dimer phase and band insulator, via Gaussian critical line can be understood as resolution (zooming) of observable over the dimer phase.





→ Almost but not exactly

$\approx$ $\neq$ $U(1) \times U(1) \longrightarrow$ "Gaussian" theory

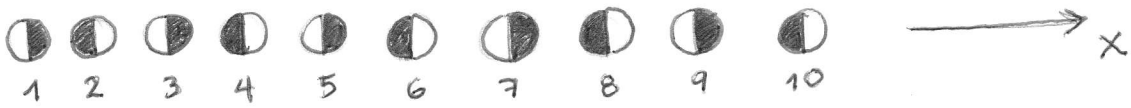
(a phase transition, seen from scale of solid)

$\not\approx$ $\neq$ $\mathbb{Z}_2 \times \mathbb{Z}_2$

$\mathbb{Z}_2$ $\mathbb{Z}_2$

→ Neither exactly nor almost

perhaps:

$\mathbb{Z}_2 \times \mathbb{1} \oplus \mathbb{1} \times \mathbb{Z}_2$

the Gaussian critical line is reached by collapsing the scale of the solids with that of the liquids.